Ali Ahmad Sabir

Backend Web Developer (Python)

ahmadultani2004@gmail.com • **+92 314 0620796** • **linkedin.com/in/aliahmadsabir-official** • **github.com/aliahmadsabir786**

PROFESSIONAL SUMMARY

Junior Python Backend Developer with hands-on experience in Backend Development, REST API Development, Authentication & Authorization, and Database Management using Python, Flask, and FastAPI. Built 2 full-scale projects — an institutional ERP (NCMH) with 40+ REST API endpoints and a Retail ERP & POS system with automated billing and Role-Based Access Control (RBAC). Strong foundation in OOP, SDLC, Debugging, and Deployment, with practical knowledge of MySQL schema design, JWT & session-based authentication, CRUD operations, and Git version control. Writes clean, modular, and maintainable code focused on solving real business problems.

TECHNICAL SKILLS

Languages: Python, JavaScript (ES6+, Basic)

Frameworks: Flask, FastAPI

Databases: MySQL, SQL, Relational Database Design, Schema Design, Query Optimization

Frontend: HTML5, CSS3, Responsive Web Design, SPA Fundamentals

API & Auth: REST API Development, JWT & Session Authentication, Role-Based Access Control (RBAC), CRUD Operations

Tools: Git, GitHub (Version Control), Postman (API Testing & Debugging), PyMySQL, Flask-Login, Werkzeug, Vercel (Deployment), Agile/Scrum

Software Engineering: Object-Oriented Programming (OOP), Backend Development, API Integration, SDLC, Debugging, Deployment, Software Testing

Concepts: Authentication & Authorization, Database Management, Data Structures, Clean Code, Modular Architecture

EXPERIENCE

Backend Developer | Academic & Personal Projects 2024 – Present

- Built backend systems using Python, Flask, and FastAPI, applying OOP principles and modular architecture for clean, maintainable code.
- Developed 40+ REST API endpoints with JWT & session-based authentication, Role-Based Access Control (RBAC), and full CRUD operations.
- Designed normalized MySQL databases with 15+ tables, wrote optimized SQL queries, and handled end-to-end SDLC including debugging, Git version control, and deployment.

PROJECTS

Nexus Campus Management Hub (NCMH) — Final Year Project 2024 – 2026

Python · Flask · FastAPI · MySQL · Flask-Login · JWT · RBAC · RESTful API · HTML5 · CSS3 · JavaScript ES6+ · SPA · Chart.js · PyMySQL · Agile

- Architected a production-ready full-stack institutional ERP with four role-based portals (Admin, Sub-Admin, Teacher, Student) using Python/Flask, RBAC decorators, Flask-Login session auth, and Werkzeug password hashing.
- Designed a normalized MySQL relational schema with 15+ tables — students, teachers, fees, exams, grades, attendance, assignments — enforcing foreign keys and referential integrity.
- Built 40+ RESTful API endpoints across modular Flask Blueprints covering full CRUD, attendance, grading, fee management, exam scheduling, and base64 file uploads.
- Engineered a vanilla JavaScript ES6+ SPA with client-side routing, Chart.js dashboards, real-time rendering, and a fully responsive mobile-first UI across 14 JS modules.

Smart Retail ERP & POS System— Full Stack Retail Management 2025 – Present

Python · Flask · FastAPI · MySQL · JWT Authentication · RBAC · REST API · HTML5 · CSS3 · JavaScript ES6 · Responsive Design · ERP · POS · Inventory Management

- Developed a full-stack ERP and POS application using Python, Flask, and FastAPI — REST APIs covering inventory management, order processing, customer management, and billing workflows.

- Designed a normalized MySQL schema for products, customers, orders, and financials, and implemented JWT authentication with RBAC for three permission-driven user roles.
- Built an automated billing engine with real-time tax/discount calculation, sale slip generation, sale/purchase return processing, and a Chart.js dashboard with CSV/PDF export.

EDUCATION

Bachelor of Science in Computer Science (BS CS) Oct 2022 – 2026 (Expected)

Government College University Faisalabad, Punjab, Pakistan

CGPA: 3.3 / 4.0 · Coursework: Data Structures, Database Systems, Software Engineering, Web Development, OOP, Networking

CERTIFICATIONS

Fundamentals of Python Programming — freeCodeCamp 2026

Introduction to Databases and SQL — Great Learning 2025